

DEFINITIONS AND FORMULAE

Area and volume formulae

Volume of a cone or pyramid	$= \frac{1}{3}Ah$, where A = base area, h = height of vertex.
Area of curved surface of a cone	$= \pi rl$, where l = slant height, r = base radius.
Volume of a sphere	$= \frac{4}{3}\pi r^3$.
Surface area of a sphere	$= 4\pi r^2$.
Area of a spherical zone	
(between planes distance h apart)	$= 2\pi rh$.

Trigonometry

$$\sec \theta = \frac{1}{\cos \theta}; \quad \csc \theta = \frac{1}{\sin \theta}; \quad \tan \theta = \frac{\sin \theta}{\cos \theta}; \quad \cot \theta = \frac{1}{\tan \theta}.$$

$$\cos^2 \theta + \sin^2 \theta = 1; \quad 1 + \tan^2 \theta = \sec^2 \theta; \quad \cot^2 \theta + 1 = \csc^2 \theta.$$

$$\sin (\theta \pm \phi) = \sin \theta \cos \phi \pm \cos \theta \sin \phi; \quad \cos (\theta \pm \phi) = \cos \theta \cos \phi \mp \sin \theta \sin \phi;$$

$$\tan (\theta \pm \phi) = \frac{\tan \theta \pm \tan \phi}{1 \mp \tan \theta \tan \phi} \quad [\theta \pm \phi \neq (k + \frac{1}{2})\pi].$$

$$\sin 2\theta = 2 \sin \theta \cos \theta; \quad \cos 2\theta = \cos^2 \theta - \sin^2 \theta; \quad \tan 2\theta = \frac{2 \tan \theta}{1 - \tan^2 \theta} \quad [\theta \neq (\frac{1}{2}k + \frac{1}{4})\pi].$$

$$2 \cos^2 \theta = 1 + \cos 2\theta; \quad 2 \sin^2 \theta = 1 - \cos 2\theta.$$

$$\text{If } t = \tan \frac{1}{2}\theta, \text{ then } \sin \theta = \frac{2t}{1+t^2}; \quad \cos \theta = \frac{1-t^2}{1+t^2}; \quad \tan \theta = \frac{2t}{1-t^2}; \quad \frac{d\theta}{dt} = \frac{2}{1+t^2}.$$

$$2 \sin \theta \cos \phi = \sin (\theta + \phi) + \sin (\theta - \phi);$$

$$2 \cos \theta \cos \phi = \cos (\theta + \phi) + \cos (\theta - \phi);$$

$$2 \sin \theta \sin \phi = \cos (\theta - \phi) - \cos (\theta + \phi).$$

$$\sin \alpha + \sin \beta = 2 \sin \frac{1}{2}(\alpha + \beta) \cos \frac{1}{2}(\alpha - \beta); \quad \sin \alpha - \sin \beta = 2 \cos \frac{1}{2}(\alpha + \beta) \sin \frac{1}{2}(\alpha - \beta);$$

$$\cos \alpha + \cos \beta = 2 \cos \frac{1}{2}(\alpha + \beta) \cos \frac{1}{2}(\alpha - \beta); \quad \cos \alpha - \cos \beta = 2 \sin \frac{1}{2}(\alpha + \beta) \sin \frac{1}{2}(\beta - \alpha).$$

In the triangle ABC :

$$\frac{a}{\sin A} = \frac{b}{\sin B} = \frac{c}{\sin C} = 2R;$$

$$a^2 = b^2 + c^2 - 2bc \cos A, \text{ etc.};$$

$$\tan \frac{1}{2}A = \sqrt{\frac{(s-b)(s-c)}{s(s-a)}}, \text{ etc.}; \quad \text{area} = \sqrt{s(s-a)(s-b)(s-c)};$$

where $s = \frac{1}{2}(a+b+c)$.

Ranges of the inverse functions:

$$-\frac{1}{2}\pi \leq \sin^{-1}x \leq \frac{1}{2}\pi; \quad 0 \leq \cos^{-1}x \leq \pi; \quad -\frac{1}{2}\pi < \tan^{-1}x < \frac{1}{2}\pi.$$

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Algebra

Series

$$\sum_{i=1}^n i = \frac{1}{2}n(n+1) \quad (\text{an arithmetic progression}).$$

$$\sum_{i=1}^n i^2 = \frac{1}{6}n(n+1)(2n+1); \quad \sum_{i=1}^n i^3 = \frac{1}{4}n^2(n+1)^2.$$

$$\sum_{i=1}^n i(i+1)(i+2)\dots(i+r) = \frac{1}{r+2} n(n+1)(n+2)\dots(n+r+1).$$

$$\sum_{i=1}^n x^{i-1} = \frac{1-x^n}{1-x} \quad (x \neq 1) \quad (\text{a geometric progression}); \quad \text{if } |x| < 1, \quad \sum_{i=1}^{\infty} x^{i-1} = \frac{1}{1-x}.$$

Binomial coefficients

If i is a natural number,

$$\binom{n}{i} = \frac{n(n-1)\dots(n-i+1)}{i!}.$$

Coefficients may be calculated from the definition

$$\binom{n}{0} = 1, \quad \binom{n}{i+1} = \frac{n-i}{i+1} \binom{n}{i}.$$

Also,

$$\binom{k+1}{i} = \binom{k}{i-1} + \binom{k}{i}.$$

If n is also a natural number, and if ${}_nC_i$ denotes the number of subsets of i elements contained in a set of n elements, then

$${}_nC_i = \binom{n}{i} = \frac{n!}{i!(n-i)!}.$$

The *binomial theorem*: If n is a natural number,

$$(b+a)^n = \sum_{i=0}^n \binom{n}{i} b^{n-i} a^i.$$

[See also p. 10.]

Logarithms, exponentials and hyperbolic functions

$$y = b^x \Leftrightarrow x = \log_b y.$$

$$y = \exp x \Leftrightarrow x = \ln y \quad (\exp x \text{ is the same as } e^x, \ln y \text{ as } \log_e y).$$

$$q^x = \exp(x \ln q).$$

$$\log_b a = \log_e a / \log_e b; \quad \text{in particular, } \ln a = \log_{10} a / \log_{10} e \approx 2.3026 \log_{10} a.$$

$$\sinh x = \frac{1}{2}(e^x - e^{-x}); \quad \cosh x = \frac{1}{2}(e^x + e^{-x}); \quad \tanh x = \sinh x / \cosh x.$$

$$\sinh^{-1} x = \ln \{x + \sqrt{(1+x^2)}\}; \quad \cosh^{-1} x = \ln \{x + \sqrt{(x^2-1)}\} \quad (x \geq 1);$$

$$\tanh^{-1} x = \frac{1}{2} \ln \{(1+x)/(1-x)\} \quad (-1 < x < 1).$$

Quadratic functions

If $a \neq 0$,

$$ax^2 + bx + c = 0 \Leftrightarrow x = \frac{-b \pm \sqrt{(b^2 - 4ac)}}{2a}.$$

Sum of roots = $-b/a$; product of roots = c/a .

In real algebra, if $a \neq 0$,

$$ax^2 + bx + c > 0 \quad \text{for all } x \Leftrightarrow a > 0 \text{ and } 4ac - b^2 > 0.$$

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Differentiation

If $\frac{f(x)-f(a)}{x-a}$ tends to a limit as x tends to a , then f is said to be differentiable at a . The limit is called the derivative of f at a and is usually written $f'(a)$. The function f' is called the derived function of f .

Derivatives of common functions

$f(x)$	$f'(x)$
x^m	mx^{m-1}
$\sin x$	$\cos x$
$\cos x$	$-\sin x$
$\tan x$	$\sec^2 x$
$\sec x$	$\sec x \tan x$
$\cot x$	$-\csc^2 x$
$\csc x$	$-\csc x \cot x$
$e^x (= \exp x)$	$e^x (= \exp x)$
a^x	$a^x \ln a$
$\ln x (= \log_e x)$	$1/x$
$\sinh x$	$\cosh x$
$\cosh x$	$\sinh x$

If $y = uv$, then

$$\frac{dy}{dx} = \frac{du}{dx} v + u \frac{dv}{dx}.$$

If $y = u/v$, then

$$\frac{dy}{dx} = \left(\frac{du}{dx} v - u \frac{dv}{dx} \right) / v^2.$$

Approximations to derivatives

If the derivatives exist, then for small h

$$f'(x) \approx \frac{f(x+h) - f(x-h)}{2h};$$

$$f''(x) \approx \frac{f(x+h) - 2f(x) + f(x-h)}{h^2}.$$

Polynomial approximations to functions

Taylor's approximation. For small h ,

$$f(a+h) \approx f(a) + f'(a) \cdot h + \frac{1}{2!} f''(a) \cdot h^2 + \dots + \frac{1}{n!} f^{(n)}(a) \cdot h^n$$

provided that the derivatives exist and are continuous at a .

Applications to particular functions:

$$(1+h)^m \approx 1 + mh + \frac{m(m-1)}{2!} h^2 + \dots + \binom{m}{n} h^n.$$

$$\ln(1+h) \approx h - \frac{h^2}{2} + \frac{h^3}{3} - \dots + (-1)^{n+1} \frac{h^n}{n}.$$

$$e^h \approx 1 + h + \frac{h^2}{2!} + \dots + \frac{h^n}{n!}.$$

$$\cosh h \approx 1 + \frac{h^2}{2!} + \frac{h^4}{4!} + \dots + \frac{h^{2k}}{(2k)!}.$$

$$\sinh h \approx h + \frac{h^3}{3!} + \dots + \frac{h^{2k+1}}{(2k+1)!}.$$

$$\cos h \approx 1 - \frac{h^2}{2!} + \frac{h^4}{4!} + \dots + (-1)^k \frac{h^{2k}}{(2k)!}.$$

$$\sin h \approx h - \frac{h^3}{3!} + \dots + (-1)^k \frac{h^{2k+1}}{(2k+1)!}.$$

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Indefinite integrals of common functions

[In the following we take $a > 0$ and omit the additive constant.]

$f(x)$	$\int f(x) dx$
$x^n (n \neq -1)$	$x^{n+1}/(n+1)$
$1/x$	$\ln x$ if $x > 0$, $\ln(-x)$ if $x < 0$ (i.e. $\ln x $, $x \neq 0$)
$\frac{1}{x^2 + a^2}$	$\frac{1}{a} \tan^{-1} \frac{x}{a}$
$\frac{1}{x^2 - a^2}$	$\frac{1}{2a} \ln \left \frac{x-a}{x+a} \right $
$\frac{1}{\sqrt{(x^2 + a^2)}}$	$\sinh^{-1} \frac{x}{a}$
$\frac{1}{\sqrt{(x^2 - a^2)}}$	$\cosh^{-1} \frac{x}{a}$ if $x > a$, $-\cosh^{-1} \left(\frac{-x}{a} \right)$ if $x < -a$
$\frac{1}{\sqrt{(a^2 - x^2)}}$	$\sin^{-1} \frac{x}{a}$
$\sin x$	$-\cos x$
$\cos x$	$\sin x$
$\tan x$	$\ln \sec x $
$\cot x$	$\ln \sin x $
$\sec x$	$\ln \sec x + \tan x = \ln \left \tan \left(\frac{1}{2}x + \frac{1}{2}\pi \right) \right $
$\csc x$	$\ln \left \tan \frac{1}{2}x \right $
$e^{ax} \sin bx$	$\frac{e^{ax}}{a^2 + b^2} (a \sin bx - b \cos bx)$
$e^{ax} \cos bx$	$\frac{e^{ax}}{a^2 + b^2} (a \cos bx + b \sin bx)$
$\sin^2 x$	$\frac{1}{2}(x - \frac{1}{2} \sin 2x)$
$\cos^2 x$	$\frac{1}{2}(x + \frac{1}{2} \sin 2x)$
$\sinh x$	$\cosh x$
$\cosh x$	$\sinh x$

For logarithmic forms of inverse hyperbolic functions see p. 3.

Integration by parts

$$\int u \frac{dv}{dx} dx = uv - \int \frac{du}{dx} v dx.$$

Reduction formulae for trigonometric integrals

$$\int_0^{\frac{1}{2}\pi} \sin^m x dx = \frac{m-1}{m} \int_0^{\frac{1}{2}\pi} \sin^{m-2} x dx; \quad \int_0^{\frac{1}{2}\pi} \cos^m x dx = \frac{m-1}{m} \int_0^{\frac{1}{2}\pi} \cos^{m-2} x dx;$$

$$\int_0^{\frac{1}{2}\pi} \sin^m x \cos^n x dx = \frac{m-1}{m+n} \int_0^{\frac{1}{2}\pi} \sin^{m-2} x \cos^n x dx = \frac{n-1}{m+n} \int_0^{\frac{1}{2}\pi} \sin^m x \cos^{n-2} x dx.$$

[These results hold provided that the exponents in the reduced form are greater than -1 . There are analogous reduction formulae with other intervals of integration $(\frac{1}{2}k_1\pi, \frac{1}{2}k_2\pi)$ with k_1, k_2 integral.]